Muhammad Haider Zaidi

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github.com/chefhaider Bavaria, Germany 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Javascript
Machine Learning: Statistics, Optimization, Deep Learning, Reinforcement Learning, Time Series Analysis
Deep Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers
MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, Real-time System Optimization, Production Deployment
Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Big Data (Hadoop Ecosystem, Hive, Sqoop, NiFi)

Experience

German National Metrology Institute (PTB) **March 2024 – Present**
Student Assistant Researcher *Berlin, DE*

- Innovated and evaluated deep learning models for metrological applications, maximizing model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability, applying state-of-the-art methods.
- Built comprehensive dashboards for real-time data visualisation and model performance monitoring, facilitating iterative insights.

Folio3 **March 2022 – Nov 2023**
Machine Learning Engineer *California, US (remote)*

- Improved vitiligo segmentation accuracy to 94% and increased inference speed by 5x using advanced AI algorithms, driving advertising performance.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and processing time by 6x, demonstrating high scalability.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure for production systems.
- Developed Proof of Concepts (POCs) using YOLO, few-shot learning, and CNN for client demos, showcasing rapid prototyping of novel algorithms.

Manda GmbH **March 2024 - June 2024**
Werkstudent Python Developer *Nürnberg, DE*

- Developed full-stack web applications providing AI-based services using Flask, HTML, CSS, enabling robust data interaction.
- Built an AI chatbot application leveraging the OpenAI API and relevant LLM tools/frameworks, demonstrating modern ML algorithm implementation.

Freelance **July 2020 – March 2022**
Data Engineer *Remote*

- Built robust web scrapers and ETL pipelines for structured and unstructured data, ensuring efficient data storage and processing.
- Automated data ingestion and cleaning workflows for client analytics use cases, supporting scalable and high-quality data for ML models.

Projects

CAFA 6: Protein Function Prediction **2022**
Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, using AlphaFold for feature extraction and taxonomic bias mitigation, enhancing model accuracy.
- Increased model accuracy by 4% through custom feature engineering across Molecular Function (MF), Cellular Component (CC), and Biological Process (BP) namespaces, showcasing optimization skills.

Razanlytics: Real-time Bitcoin Price Forecasting **2022**
Thesis Project

- Developed a real-time Bitcoin price forecasting system, applying advanced statistical and machine learning models for predictive accuracy and advertising performance optimization.
- Implemented comprehensive data pipelines for historical data ingestion and processing, crucial for iterative model testing and insights.

Saaz: AI Self-Playing Chitrali Sitar **2023**
Signal Processing, Music Generation

- Designed a real-time signal processing system integrating temporal dependencies for music generation, demonstrating complex algorithm innovation.
- Implemented transformer-based models for sequence prediction in audio generation tasks, applying state-of-the-art deep learning methods.

Hugging Face **Open-Source**
Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, showcasing understanding of state-of-the-art ML methods.

Education

Friedrich Alexander Universität **Present**
Master of Science in Artificial Intelligence *Erlangen, DE*

- Specialized in deep learning for computer vision and spatio-temporal modeling, relevant for ROI maximization and advertising performance.
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems, focusing on cutting-edge ML algorithms.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**
Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis, providing a comprehensive understanding of ML and statistics.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera**: TensorFlow Developer Certificate | **Databricks**: Lakehouse Fundamentals | **Datacamp**: Computer Vision in Python